

Biological Anthropology

Question & Answer Review — Complete Answer Key (50 Questions)

1. Fossil found with inferior foramen magnum position, valgus knee, and non-honing canines — is this a hominin fossil?

Yes / No

Answer: Yes

All three traits (anterior foramen magnum, valgus knee, non-honing canines) are classic diagnostic markers of bipedalism and the hominin lineage.

2. What species in the genus *Homo* retains *Australopithecine* traits but is relatively recent?

Answer: *Homo naledi*

Small brain (~460–610 cc) and other primitive traits, yet dated to only ~236,000–335,000 years ago (Dinaledi Chamber, South Africa), overlapping with early *Homo sapiens*.

3. True or False: There is no overlap in time between hominin species.

True / False

Answer: False

Many hominin species coexisted in time (e.g., *H. naledi* with *H. sapiens*; *H. floresiensis* with *H. sapiens* and *H. erectus*; multiple *Australopithecus*/*Paranthropus* species with early *Homo*).

4. True or False: Apes evolved in Africa and then migrated to Asia before Europe.

True / False

Answer: False

Miocene apes dispersed Africa → Europe → Asia. *Griphopithecus* appears in Europe/Anatolia by ~16.5 mya, while *Sivapithecus* doesn't appear in Asia until ~12.5 mya.

5. Which of the following species is considered a robust *Australopithecine*?

sediba / *garhi* / *afarensis* / *robustus*

Answer: *robustus*

A. (Paranthropus) robustus has heavy craniodental adaptations (sagittal crest, large posterior teeth). *Sediba*, *garhi*, and *afarensis* are gracile forms.

6. Which of the following hominin has partial canine honing? (1st ask)

Sahelanthropus tchadensis / *Ardipithecus kadabba* / *Ardipithecus ramidus* / *Homo habilis*

Answer: *Ardipithecus kadabba*

Ar. kadabba (~5.8–5.2 mya) shows a transitional, partially honing canine/premolar complex, more primitive than the fully non-honing condition in *Ar. ramidus* and later hominins.

7. During which epoch did apes emerge?

Answer: Miocene

The Miocene (~23–5.3 mya) is known as the “Age of Apes,” marked by extensive hominoid radiation across Africa, Europe, and Asia.

8. What muscles are associated with chewing?

Answer: Temporalis, masseter, medial pterygoid, lateral pterygoid

The temporalis and masseter are the main “power” muscles; their attachment sites (sagittal crest, zygomatic flaring) are key diagnostic features in the fossil record.

9. True or False: *Homo habilis* had the same average height as modern *Homo sapiens*.

True / False

Answer: False

H. habilis stood roughly 3.5–4.5 ft tall, much shorter than modern *H. sapiens*; modern-like stature didn't appear until *H. erectus*.

10. Which of the following is a possible ancestor to primates?

Sciuravids / Plesiadapiforms / (Aegyptopithecus) / Adapids

Answer: Plesiadapiforms

Paleocene archaic mammals considered the likely stem group/ancestors of true primates. Adapids are themselves true primates (ancestral to strepsirrhines), not ancestors of primates generally.

11. What species of fossil hominin is thought to be a direct descendant of *Ardipithecus kadabba*?

Answer: *Ardipithecus ramidus*

Ar. kadabba (~5.8–5.2 mya) is considered ancestral to *Ar. ramidus* (~4.4 mya); the two are sometimes treated as a single chronospecies lineage.

12. Which of the following sites has footprints of *Homo erectus*?

Holyoke / Ileret / Le Rozel / Laetoli

Answer: Ileret

Ileret, Kenya preserves *H. erectus* trackways (~1.5 mya) showing a modern human-like gait. Laetoli = *A. afarensis* footprints; Le Rozel = Neanderthal footprints; Holyoke = dinosaur tracks.

13. What is the time it takes for half of an isotope to decay called?

Answer: Half-life

Basis of radiometric dating methods like K-Ar/Ar-Ar (older fossils) and radiocarbon dating (younger material, up to ~50,000 years).

14. True or False: The visual predation hypothesis states that the first primate evolved because it was adapting to hunting bugs and small prey in trees.

True / False

Answer: True

Cartmill's hypothesis: forward-facing eyes and grasping hands/feet evolved for visually-guided predation on insects/small prey, not simply for arboreal locomotion generally.

15. True or False: Early apes retained quadrupedal locomotion.

True / False

Answer: True

Early Miocene apes like Proconsul moved via above-branch quadrupedalism, monkey-like, rather than the specialized suspensory locomotion of later apes.

16. True or False: Modern Homo sapiens have smaller browridges than archaic Homo sapiens.

True / False

Answer: True

Reflects the broader gracilization trend from archaic to modern H. sapiens: smaller brows, more vertical forehead, thinner cranial bone, presence of a chin.

17. Which of the following traits is associated with bipedalism?

C-shaped spine / Valgus knee / Opposable big toe / Posterior foramen magnum position

Answer: Valgus knee

Angled femur keeps knees under the body's midline for a bipedal stride. The other three options are traits of quadrupedal/arboreal locomotion instead.

18. Brain size expansion within the hominin line is associated with which genus?

Answer: Homo

Cranial capacity rises sharply across genus Homo: H. habilis (~510–650cc) → H. erectus (~700–1100cc) → H. heidelbergensis/neanderthalensis/sapiens (~1200–1750cc).

19. Which of the following species is sometimes split into two, with Homo rudolfensis?

naledi / habilis / floresiensis / erectus

Answer: Homo habilis

The variable H. habilis fossil sample is sometimes split into smaller-brained H. habilis (sensu stricto) and larger-brained, larger-faced H. rudolfensis (e.g., KNM-ER 1470).

20. What is the smallest member of the genus Homo?

naledi / erectus / sapiens / floresiensis

Answer: Homo floresiensis

"The Hobbit" stood ~3.5 ft tall with a brain of ~380–420cc, likely via island dwarfism. H. erectus was actually one of the taller early Homo species (e.g., Turkana Boy, ~6 ft).

21. Which of the following features is associated with robust Australopithecines/Paranthropines?

Sagittal crest / Small posterior teeth / Large brain / Honing canines

Answer: Sagittal crest

Anchors an enlarged temporalis muscle for heavy chewing. Robust australopiths actually have large posterior teeth, small brains, and non-honing canines—the opposite of the other three options.

22. True or False: Fossils are the remains of organisms chemically changed into rock.

True / False

Answer: False

Too narrow a definition—fossils also include trace fossils (footprints), molds/casts, and unaltered remains (permafrost, amber), not just permineralized/rock-like bone.

23. Which of the following trends occur over time in genus Homo?

All of the options / Brain size expansion / Reduction in tooth size

Answer: All of the options

Both brain expansion and tooth-size reduction are well-documented trends across Homo evolution, part of the broader “gracilization” pattern.

24. What species of Australopithecus is the Lucy fossil?

Answer: Australopithecus afarensis

AL 288-1, found at Hadar, Ethiopia (1974), dated to ~3.2 mya; shows a mix of bipedal adaptations and primitive, ape-like traits.

25. Which tool type is associated with Homo erectus? (1st ask)

Answer: Acheulean

Bifacial hand axes and cleavers, a major advance over the earlier, simpler Oldowan tools associated with H. habilis.

26. What hypothesis for the emergence of hominins is associated with Charles Darwin? (Options: hunting, provisioning, patchy forest, savannah)

hunting hypothesis / provisioning hypothesis / patchy forest hypothesis / savannah hypothesis

Answer: Hunting hypothesis

Darwin proposed bipedalism, tool/weapon use (for hunting), canine reduction, and brain expansion evolved together as an interconnected package.

27. Which of the following traits is not associated with Homo erectus?

Sagittal crest / Long and low skull / Large browridge / Thick cranial bones

Answer: Sagittal crest

A true sagittal crest belongs to robust australopithecines; H. erectus may show only a subtle sagittal keel, a much less pronounced feature.

28. True or False: Fossils are well preserved at all locations around the globe.

True / False

Answer: False

Preservation depends heavily on sediment type, burial rate, climate, and geological stability—explaining why hominin fossils cluster in places like the East African Rift Valley and South African caves.

29. What species of fossil hominin is thought to be the ancestor to Australopithecus afarensis? (1st ask)

Answer: Australopithecus anamensis

Considered a direct ancestor (~4.2–3.9 mya); often treated as part of a single anagenetic lineage leading into *A. afarensis*.

30. What is the study of the processes that occur to an organism's remains after death?

Answer: Taphonomy

Covers decomposition, scavenging, weathering, burial, transport, and fossilization—key to distinguishing genuine behavioral evidence from post-mortem alteration.

31. What model for the origin of modern *Homo sapiens* argues that modern humans evolved in Africa and there was no interbreeding with archaic populations?

Answer: Replacement Model (“Out of Africa” / Recent African Origin, strict version)

Contrasts with the Multiregional Continuity Model (simultaneous evolution across regions) and the Assimilation Model (African origin + some interbreeding, supported by modern Neanderthal-DNA evidence).

32. How many waves of migration to the North American continent took place prior to contact with Columbus?

Answer: Three

Paleoindians (ancestors of most Native Americans), Na-Dene speakers (e.g., Athabaskan groups), and Inuit-Aleut (Eskimo-Aleut) populations.

33. Does *Homo naledi* have a larger brain than *Homo habilis*?

Answer: No

H. naledi: ~460–610cc; *H. habilis*: ~510–650cc. Despite *H. naledi* being far more recent, its brain size is comparable to or smaller than *H. habilis*—a hallmark of its mosaic morphology.

34. True or False: Modern *Homo sapiens* are descended from *Homo naledi*.

True / False

Answer: False

H. naledi overlapped in time with early *H. sapiens* rather than preceding them; it represents a separate, late-surviving side branch, not a direct ancestor.

35. What is the earliest South African hominin?

Answer: “Little Foot” (*Australopithecus prometheus* / *africanus*)

Found at Sterkfontein, dated to ~3.67 mya—older than most *A. afarensis* specimens. (*A. africanus*, via the Taung Child, is the classic “first species described” answer, dated ~2.1–2.6 mya.)

36. Which of the following sites is associated with *Australopithecus afarensis*?

Daka, Ethiopia / Makapansgat, South Africa / Bouri, Ethiopia / Hadar, Ethiopia

Answer: Hadar, Ethiopia

Type site for *A. afarensis* (including Lucy). Makapansgat = *A. africanus*; Bouri = *A. garhi*; Daka = *H. erectus*.

37. What is the earliest type of stone tool?

Answer: Oldowan

Simple flakes and choppers dating to ~2.6 mya, named for Olduvai Gorge; associated primarily with early Homo (traditionally H. habilis).

38. What is the oldest species of fossil hominin?

Answer: Sahelanthropus tchadensis

Dated to ~7 mya (Chad), around the estimated human-chimpanzee divergence; already shows small, non-honing canines and an anterior foramen magnum.

39. Which of the following is a disadvantage of bipedalism?

Decreased ability to see distances / Hands occupied with carrying things / Can only run short distances / Greater exposure to predators

Answer: Greater exposure to predators

Slower locomotion and increased visibility while upright increase predation risk. (The other options are actually reversed—increased visual range and free hands are advantages of bipedalism.)

40. True or False: Homo habilis was the first hominin species to live in Europe.

True / False

Answer: False

H. habilis never left Africa. Homo erectus was the first hominin to disperse out of Africa into Europe and Asia (e.g., Dmanisi, Georgia; Atapuerca, Spain).

41. Which species is believed to have been the first to cook its food?

Answer: Homo erectus

Per Wrangham's "cooking hypothesis": control of fire and cooking allowed smaller teeth/guts and more calories to fuel brain growth. Possible early fire evidence at Koobi Fora and Swartkrans (~1.5 mya).

42. The earliest true primate emerged during what geological epoch?

Answer: Eocene

True primates (Euprimates)—Adapiforms and Omomyiforms—show the full suite of primate traits (postorbital bar, forward-facing eyes, nails), distinct from the preceding Paleocene Plesiadapiforms.

43. Biostratigraphy dating problem: Pig species A (2–1 mya), B (2–1.5 mya), C (1.75–0.5 mya) all present at an East African site—what is the estimated site age?

Answer: 1.5 to 1.75 mya

Biostratigraphy uses the overlap of all species' date ranges: latest "start" = 1.75 mya (species C); earliest "end" still shared by all = 1.5 mya (species B).

44. What is the first "tall" hominin? (Options: naledi, habilis, sapiens, erectus)

Homo naledi / Homo habilis / Homo sapiens / Homo erectus

Answer: Homo erectus

First Homo species with modern human-like stature (e.g., Turkana Boy, ~6 ft), a major increase over smaller-bodied earlier Homo species.

45. True or False: There are no fossils of Homo erectus in Asia.

True / False

Answer: False

Major Asian *H. erectus* finds include “Java Man” (Trinil, Indonesia) and “Peking Man” (Zhoukoudian, China).

46. Which of the following is an absolute dating method?

Stratigraphic Correlation / Radiocarbon Dating / Fluorine Dating / Biostratigraphy

Answer: Radiocarbon Dating

Chronometric method giving a specific numerical age via isotope decay/half-life. The other three are relative dating methods.

47. Which of the following is not a hominin?

Australopithecus boisei / Sahelanthropus tchadensis / Sivapithecus indicus / Homo erectus

Answer: Sivapithecus indicus

A Miocene ape (~12.5 mya, India), likely ancestor to orangutans—not part of the hominin lineage. The other three are all hominins.

48. What species of fossil hominin is thought to be the ancestor to *Australopithecus afarensis*? (2nd ask)

Answer: *Australopithecus anamensis*

Same as above (Q29)—direct ancestor, often considered part of a single anagenetic lineage with *A. afarensis*.

49. Which of the following hominin has partial canine honing? (2nd ask)

Ardipithecus kadabba / Ardipithecus ramidus / Sahelanthropus tchadensis / Homo habilis

Answer: *Ardipithecus kadabba*

Same as above (Q6)—transitional, partially honing canine/premolar complex.

50. Which tool type is associated with *Homo erectus*? (2nd ask)

Answer: Acheulean

Same as above (Q25)—bifacial hand axes and cleavers.